Production Efficiency Intelligence

Synthetic manufacturing data · API-calculated OEE · AI insight for human review

Lines

Packaging Line 1 (line-1) — ideal cycle 30s
Filling Line 2 (line-2) — ideal cycle 24s

Inspect a simulated shift

Run the simulator in a terminal and paste the printed shift ID below.

Shift ID

4907d3b0-107a-4f47-8463-c972db68b658

Load OEE, losses, events, and insight

OEE API Losses API Events API

OEE


```
{
  "shift": {
    "id": "4907d3b0-107a-4f47-8463-c972db68b658",
    "shiftName": "shift-line-1-20261106-010147",
```

```
    "lineId": "line-1",
    "startUtc": "2026-11-06T01:01:47.050704+00:00",
    "endUtc": "2026-11-06T09:01:47.050704+00:00",
    "plannedProductionSeconds": 28800
  },
  "metrics": {
    "availabilityPercent": 89.38,
    "performancePercent": 102.21,
    "qualityPercent": 98.06,
    "oeePercent": 89.58,
    "plannedProductionMinutes": 480,
    "downtimeMinutes": 51,
    "runTimeMinutes": 429,
    "totalUnits": 877,
    "goodUnits": 860,
    "rejectedUnits": 17,
    "idealCycleTimeSeconds": 30
  }
}
```


Losses


```
{
  "shiftId": "4907d3b0-107a-4f47-8463-c972db68b658",
  "shiftName": "shift-line-1-20261106-010147",
  "losses": {
    "totalDowntimeMinutes": 51,
    "topDrivers": [
      {
        "reason": "Product changeover",
        "occurrences": 1,
        "durationMinutes": 20,
        "evidenceEventIds": [
          "d83b5aa7-72fb-4928-84f9-640e988205ab"
        ]
      },
      {
        "reason": "Minor jam",
        "occurrences": 2,
        "durationMinutes": 15,
        "evidenceEventIds": [
```

```
          "c2d41615-001f-406d-812b-fb32fa0c38a3",
          "beb73556-8a47-4618-8b7b-164e4ff09bfd"
        ]
      },
      {
        "reason": "Material wait",
        "occurrences": 1,
        "durationMinutes": 10,
        "evidenceEventIds": [
          "b36c76e7-4311-44bd-a1ff-c6b574bb58c5"
        ]
      },
      {
        "reason": "Quality inspection",
        "occurrences": 1,
        "durationMinutes": 6,
        "evidenceEventIds": [
          "9bdf5b74-5cb4-4418-96e0-56d17242b86f"
        ]
      }
    ]
  }
}
```


Events


```
[
  {
    "id": "32f64730-94de-4e83-b8b2-a55b652222ee",
    "shiftId": "4907d3b0-107a-4f47-8463-c972db68b658",
    "type": "Production",
    "timestampUtc": "2026-11-06T01:16:47.050704+00:00",
    "startedAtUtc": null,
    "endedAtUtc": null,
    "unitsProduced": 90,
    "rejectedUnits": 2,
    "reason": null
  },
  {
    "id": "6833b135-f445-40ed-b56e-2a3c87f7d2de",
    "shiftId": "4907d3b0-107a-4f47-8463-c972db68b658",
```

```
    "type": "Production",
    "timestampUtc": "2026-11-06T01:56:47.050704+00:00",
    "startedAtUtc": null,
    "endedAtUtc": null,
    "unitsProduced": 88,
    "rejectedUnits": 1,
    "reason": null
  },
  {
    "id": "200b233c-ffdd-4f7d-a5cc-05622597b586",
    "shiftId": "4907d3b0-107a-4f47-8463-c972db68b658",
    "type": "Production",
    "timestampUtc": "2026-11-06T02:36:47.050704+00:00",
    "startedAtUtc": null,
    "endedAtUtc": null,
    "unitsProduced": 84,
    "rejectedUnits": 2,
    "reason": null
  },
  {
    "id": "c2d41615-001f-406d-812b-fb32fa0c38a3",
    "shiftId": "4907d3b0-107a-4f47-8463-c972db68b658",
    "type": "Downtime",
    "timestampUtc": "2026-11-06T02:41:47.050704+00:00",
    "startedAtUtc": "2026-11-06T02:41:47.050704+00:00",
    "endedAtUtc": "2026-11-06T02:49:47.050704+00:00",
    "unitsProduced": 0,
    "rejectedUnits": 0,
    "reason": "Minor jam"
  },
  {
    "id": "ab581998-d7bd-46b5-9b07-0284b2693cda",
    "shiftId": "4907d3b0-107a-4f47-8463-c972db68b658",
    "type": "Production",
    "timestampUtc": "2026-11-06T03:16:47.050704+00:00",
    "startedAtUtc": null,
    "endedAtUtc": null,
    "unitsProduced": 86,
    "rejectedUnits": 1,
    "reason": null
  },
  {
    "id": "beb73556-8a47-4618-8b7b-164e4ff09bfd",
    "shiftId": "4907d3b0-107a-4f47-8463-c972db68b658",
    "type": "Downtime",
    "timestampUtc": "2026-11-06T03:21:47.050704+00:00",
    "startedAtUtc": "2026-11-06T03:21:47.050704+00:00",
```

```
    "endedAtUtc": "2026-11-06T03:28:47.050704+00:00",
    "unitsProduced": 0,
    "rejectedUnits": 0,
    "reason": "Minor jam"
  },
  {
    "id": "dd3aa266-5f0a-4223-ba83-9202901d08c7",
    "shiftId": "4907d3b0-107a-4f47-8463-c972db68b658",
    "type": "Production",
    "timestampUtc": "2026-11-06T03:56:47.050704+00:00",
    "startedAtUtc": null,
    "endedAtUtc": null,
    "unitsProduced": 87,
    "rejectedUnits": 2,
    "reason": null
  },
  {
    "id": "62a73dd2-46dc-471d-a6c9-21859f87fcb1",
    "shiftId": "4907d3b0-107a-4f47-8463-c972db68b658",
    "type": "Production",
    "timestampUtc": "2026-11-06T04:36:47.050704+00:00",
    "startedAtUtc": null,
    "endedAtUtc": null,
    "unitsProduced": 89,
    "rejectedUnits": 2,
    "reason": null
  },
  {
    "id": "d83b5aa7-72fb-4928-84f9-640e988205ab",
    "shiftId": "4907d3b0-107a-4f47-8463-c972db68b658",
    "type": "Changeover",
    "timestampUtc": "2026-11-06T04:56:47.050704+00:00",
    "startedAtUtc": "2026-11-06T04:56:47.050704+00:00",
    "endedAtUtc": "2026-11-06T05:16:47.050704+00:00",
    "unitsProduced": 0,
    "rejectedUnits": 0,
    "reason": "Product changeover"
  },
  {
    "id": "7c09151d-cecd-4996-a505-5b9379be779f",
    "shiftId": "4907d3b0-107a-4f47-8463-c972db68b658",
    "type": "Production",
    "timestampUtc": "2026-11-06T05:16:47.050704+00:00",
    "startedAtUtc": null,
    "endedAtUtc": null,
    "unitsProduced": 88,
    "rejectedUnits": 2,
```

```
    "reason": null
  },
  {
    "id": "94501018-378a-47f0-9bbb-60338db68f3f",
    "shiftId": "4907d3b0-107a-4f47-8463-c972db68b658",
    "type": "Production",
    "timestampUtc": "2026-11-06T05:56:47.050704+00:00",
    "startedAtUtc": null,
    "endedAtUtc": null,
    "unitsProduced": 88,
    "rejectedUnits": 2,
    "reason": null
  },
  {
    "id": "d1528159-d6bf-4e57-a52f-e6fea54e655a",
    "shiftId": "4907d3b0-107a-4f47-8463-c972db68b658",
    "type": "Production",
    "timestampUtc": "2026-11-06T06:36:47.050704+00:00",
    "startedAtUtc": null,
    "endedAtUtc": null,
    "unitsProduced": 91,
    "rejectedUnits": 1,
    "reason": null
  },
  {
    "id": "b36c76e7-4311-44bd-a1ff-c6b574bb58c5",
    "shiftId": "4907d3b0-107a-4f47-8463-c972db68b658",
    "type": "Downtime",
    "timestampUtc": "2026-11-06T06:51:47.050704+00:00",
    "startedAtUtc": "2026-11-06T06:51:47.050704+00:00",
    "endedAtUtc": "2026-11-06T07:01:47.050704+00:00",
    "unitsProduced": 0,
    "rejectedUnits": 0,
    "reason": "Material wait"
  },
  {
    "id": "c2576c20-28c0-4793-8f18-225a50083782",
    "shiftId": "4907d3b0-107a-4f47-8463-c972db68b658",
    "type": "Production",
    "timestampUtc": "2026-11-06T07:16:47.050704+00:00",
    "startedAtUtc": null,
    "endedAtUtc": null,
    "unitsProduced": 86,
    "rejectedUnits": 2,
    "reason": null
  },
  {
```

```
    "id": "9bdf5b74-5cb4-4418-96e0-56d17242b86f",
    "shiftId": "4907d3b0-107a-4f47-8463-c972db68b658",
    "type": "Downtime",
    "timestampUtc": "2026-11-06T07:41:47.050704+00:00",
    "startedAtUtc": "2026-11-06T07:41:47.050704+00:00",
    "endedAtUtc": "2026-11-06T07:47:47.050704+00:00",
    "unitsProduced": 0,
    "rejectedUnits": 0,
    "reason": "Quality inspection"
  }
]
```


AI insight


```
{
  "shiftId": "4907d3b0-107a-4f47-8463-c972db68b658",
  "shiftName": "shift-line-1-20261106-010147",
  "provider": "Ollama",
  "model": "llama3.2",
  "generatedAtUtc": "2026-11-06T09:31:54.4146032+00:00",
  "summary": "OEE: 89.58%, Loss: 51 downtime minutes, Top drivers: Product changeover (1 occurrence), Minor jam (3 occurrences),
Material wait (1 occurrence), Quality inspection (1 occurrence)",
  "evidenceEventIds": [
    "d83b5aa7-72fb-4928-84f9-640e988205ab",
    "c2d41615-001f-406d-812b-fb32fa0c38a3",
    "beb73556-8a47-4618-8b7b-164e4ff09bfd",
    "b36c76e7-4311-44bd-a1ff-c6b574bb58c5",
    "9bdf5b74-5cb4-4418-96e0-56d17242b86f"
  ],
  "recommendations": [
    {
      "text": "Analyze the cause of the minor jam and implement corrective actions to reduce downtime.",
      "evidenceEventIds": [
        "c2d41615-001f-406d-812b-fb32fa0c38a3",
        "beb73556-8a47-4618-8b7b-164e4ff09bfd"
      ]
    },
    {
      "text": "Optimize material handling processes to minimize material wait times.",
      "evidenceEventIds": [
        "b36c76e7-4311-44bd-a1ff-c6b574bb58c5"
```

```
        ]
      }
    ]
  }
```